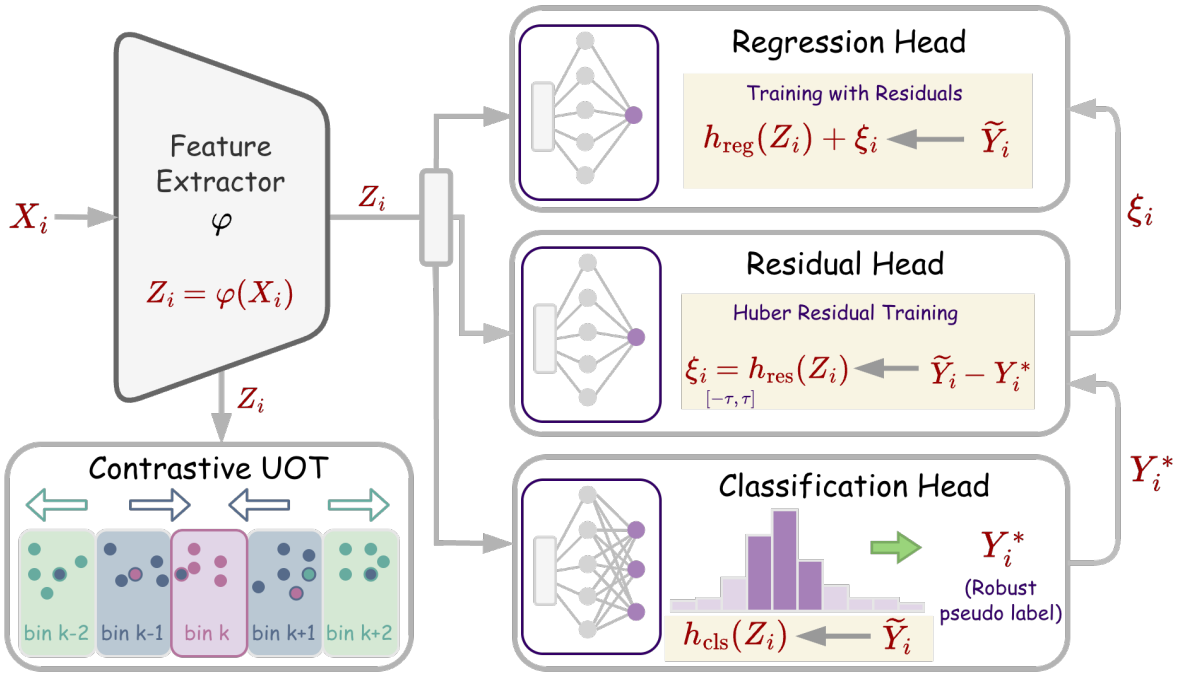

Figure 1: Overview of the HUBOT training framework. The feature extractor φ maps input X_i to shared feature Z_i, passed to three heads. **Classification head**: maps Z_i to bin-wise logits via cross-entropy loss against $\widetilde{Y}_i$, and constructs robust pseudo label Y_i^* from top-V softmax predictions. **Residual head**: learns instance-specific *within-τ residual* ξ_i, trained against $\widetilde{Y}_i - Y_i^*$, explicitly decoupling label noise from the predictive signal. **Regression head**: produces prediction $h_{\text{reg}}(Z_i)$, trained against $\widetilde{Y}_i - \xi_i$. The feature extractor is additionally regularized by **contrastive UOT loss**: features from anchor bin k (■) are aligned with nearby bins (positive pairs, ■) and separated from distant bins (negative pairs, ■), with outlier mass downweighted via marginal relaxation, providing structural protection against *label noise beyond τ across bins*, and enforcing an *order-preserving* feature geometry.